

Spring 2016 Graduating Senior - Post Graduate Contact Information and or Permanent Address

Randomized Receipt Code for Senior Exit Survey	16165
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http://www.uic-che.org/assessment/students/seniors_2016_spring/survey_exit.php

First Name: Hector
Last Name: Castaneda
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State: Illinois
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What is your preferred method of contact from the Chemical Engineering Department?

- ☒ Email
☐ Home Address
☐ Home/Work/Cellular Phone
☐ Other:

Career placement after graduation (company, graduate school, government agency, etc.)

Company / institution / agency:
Division / department / facility: Technical Sales
City and state: Chicago / Illinois
Nature of work: No career selected yet, however this will be my focus.

**UIC Department of Chemical Engineering
Anonymous Senior Exit Survey
Spring 2016**

Your survey responses has been successfully processed and recorded. Thank you for your feedback.

PLEASE WRITE DOWN THIS RECEIPT CODE: 16165

BSCHE Graduate Written Examination – Spring 2016

Written examination conducted with graduating seniors in Chemical Engineering

Name: Hector Costareda

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

In 5-10 years I would like to be working in a technical sales field with a steady company, slowly working my way up the hierarchy ladder. I am confident with my sales experience I can successfully stick with a company that sells engineering products.

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

I have researched career plans and narrowed my choices to R&D, technical sales, & design. Technical sales would be my main focus & have talked to multiple companies searching for engineers in this field.

Does your prospective employer sponsor master's degree or MBA? Under which terms?

The ones I have researched, no. They are smaller companies that do not have the capacity to sponsor.

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Currently there is a huge debate on the elimination of Fracturation, which is used in the oil and gas industry. This is evidenced through the opinions of the 2016 election. There are many people who want Fracturation completely banned, despite how dependent we are on it. Fracturation does impact the environment greatly, but I believe there is a way it can be modified so we can keep the technology but reduce harsh effects. Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

I do feel my openness with my sexuality has proved to hinder my ability to create close bonds with the students in my field of study, particularly the men. There have been instances they make comments or feel my work is inadequate giving no reason why despite the accuracy and thoroughness of my work.

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

So far I have attacked ethical dilemmas with gut reaction & personal experience; knowing what does and doesn't work. In the event I feel I cannot attack a problem alone, I go to someone for guidance such as a professor or a Human Resources representative.

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

I am familiar with the AIChE code of Ethics. A specific ~~code~~ ^{code} is to advise employers & clients if a member feels a consequence of their duties will affect the present, or future health or safety of colleague or public. A provision is illustrated in their sexual harassment policies where misconduct can lead to counseling & disciplinary action for the offender.

Which courses at UIC have contributed most to your views on professional ethics? How?

Courses that contributed to my views on professional ethics include communications, psychology, & sociology courses. They taught me how to effectively handle situations in which there is an ethical conflict through understanding and proper communication.

Question 3**Working in multidisciplinary teams**

Can you name two other types of engineers that you might find yourself working with?

*I see myself working with Industrial or Mechanical engineers.
(Aside from other chemical engineers)*

In what context or application would you be collaborating with them?

Industrial engineers to optimize complex processes, and mechanical engineers since they would have a huge contribution to the maintenance & design of machinery. Both will help make effective solutions.

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or Maple):
choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} [x_{\text{in}}(t) - x(t)] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

6) Applying the Trapezoid Rule on excel to Find the area under the curve. See Attached excel sheet.

$$F(x) = \frac{1}{1 + \cos^3 x + \ln x}$$

* plug in values From $\frac{1}{e}$ to $\frac{7}{8}$ and Find area under the curve From 1-7 using the Trapezoid Rule.

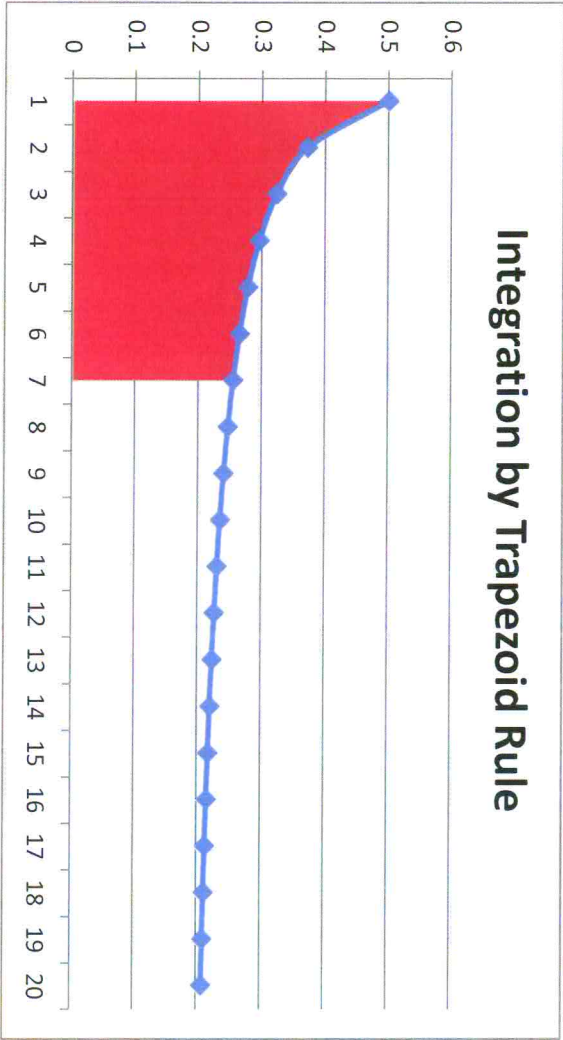
$$\int_a^b F(x) dx \approx \frac{h}{2} \sum_{k=1}^N (F(x_{k+1}) + F(x_k))$$

Based on Excel evaluation:

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x} \approx \boxed{11.52}$$

Integration by Trapezoid Rule

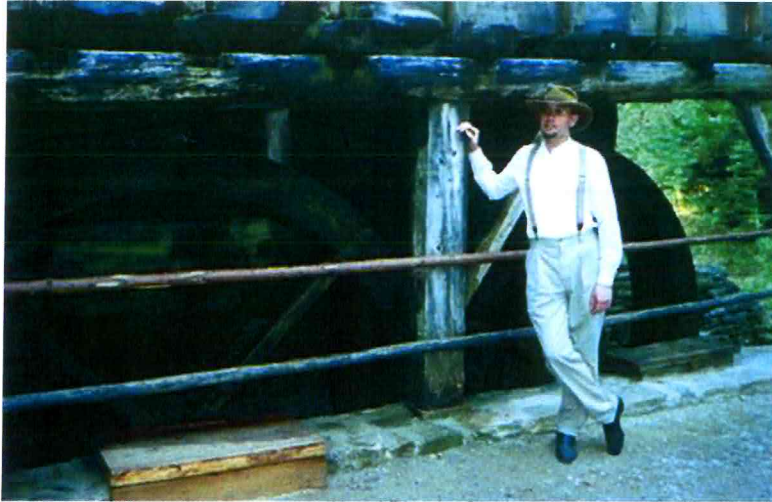
x	y	Definite Integral Area	Trapezoid Rule Equation	Summation
1	0.500114	0.500114237	0.750057119	11.51678808
2	0.371565	0.371564776	1.185782388	
3	0.323153	0.323153302	1.661576651	
4	0.295945	0.29594517	2.147972585	
5	0.277927	0.277927133	2.638963567	3.132435771
6	0.264872	0.264871543	3.132435771	
7	0.254861	0.254860517		
8	0.246881			
9	0.240342			
10	0.234869			
11	0.230214			
12	0.226205			
13	0.222716			
14	0.219656			
15	0.216954			
16	0.214555			
17	0.212415			
18	0.210499			
19	0.20878			
20	0.207232			



Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?



Overshot Water Wheel

Assumptions: Radius of the wheel is approximately height of the man ≈ 5.5 Ft

$$\Rightarrow \text{Circumference} = 2\pi r = (2)(\pi)(5.5 \text{ Ft}) = 34.56 \text{ Ft/revolution}$$

- Efficiency of water wheels is usually 67-90%. Let's assume an efficiency of 70% because it appears flow will be slowed due to the age & condition of these wheels. Lots of corrosion & appears to be made of wood.

$$* \frac{\cancel{\text{Flowrate}} (\eta) (\text{Flowrate})}{\text{Circumference}} = \frac{1 \text{ revolution}}{15 \text{ seconds}} = 4 \text{ rev/min}$$

$$\text{Flowrate} = (4 \text{ rev/min})(34.56 \text{ Ft/rev})(.70) = 197.47 \text{ Ft/min}$$

$$\text{Power (HP)} = (\text{constant})(Q)(\text{Height})(\eta)$$

$$\text{constant} = (\text{water weight lbs.}) / (\text{Foot-lbs/sec}) = 0.113491$$

$$Q = \text{Quantity of water} = (\text{FlowRate})(\text{Area})$$

* Assume 1 Foot deep & 2 Feet wide buckets

$$Q = (197.47 \text{ Ft/min})(1 \text{ Foot})(2 \text{ Ft}) = 394.94 \frac{\text{Ft}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ sec}} = 6.58 \frac{\text{Ft}^3}{\text{s}}$$

$$H \approx (2)(5.5 \text{ Ft}) = 11 \text{ Feet}$$

$$\eta = 70\%$$

$$\text{Power} = (0.113491)(6.58 \frac{\text{Ft}^3}{\text{s}})(11 \text{ Ft})(.70)$$

$$= 5.74 \text{ horse power}$$

$\Rightarrow$ This wheel produces approximately 5.74 hp

A typical home needs 4,000 - 7,000 watts of power to function, or 5.74 hp = 4286 watts. This picture shows multiple wheels, so more than one is used to produce a sufficient amount of power.